

Data Structures

CSE207

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Stack

- It is a linear data structure consisting of list of items.
- In stack, data elements are **added or removed** only at one end, called ***the top of the stack***.

Stack: AAA, BBB, CCC, DDD, EEE

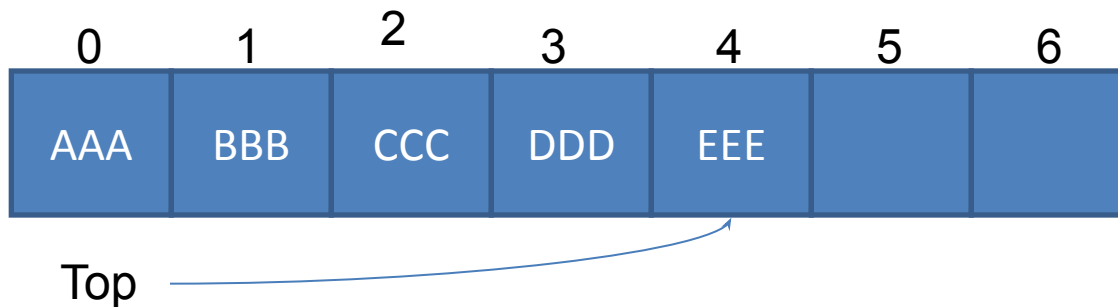
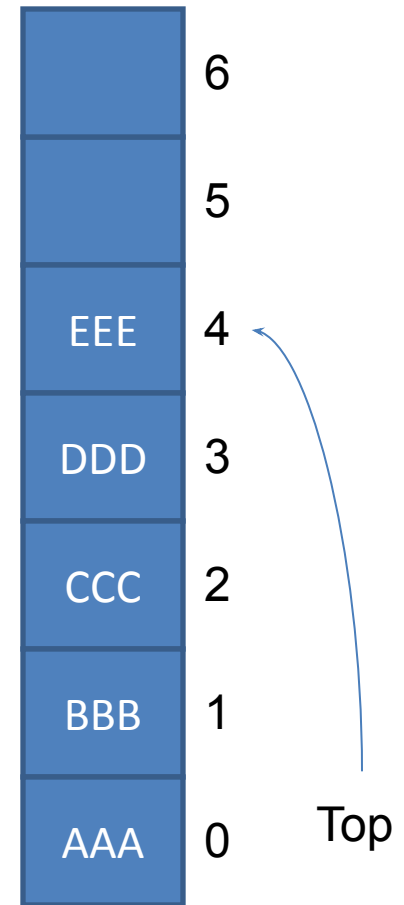


Figure: Diagrams of Stacks

Stack

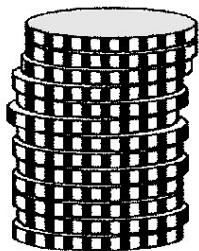
- Two basic operations are associated with stack:
- “Push” operation is used to insert an element into a stack.
- “Pop” operation is used to delete an element from a stack.

Stack: AAA, BBB, CCC, DDD, EEE

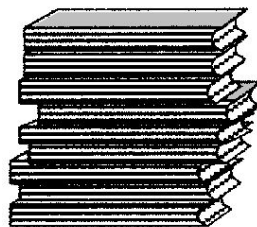


Stack

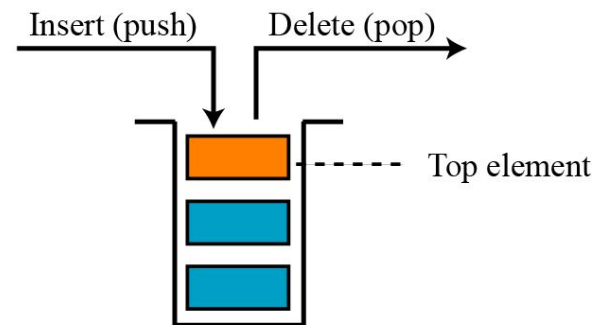
- A stack is a restricted linear list in which all additions and deletions are made at one end, the top.
- If we insert a series of data items into a stack and then remove them, the order of the data is reversed.
- This reversing attribute is why stacks are known as last in, **first out (LIFO) data structures**.



Stack of coins



Stack of books



Computer stack

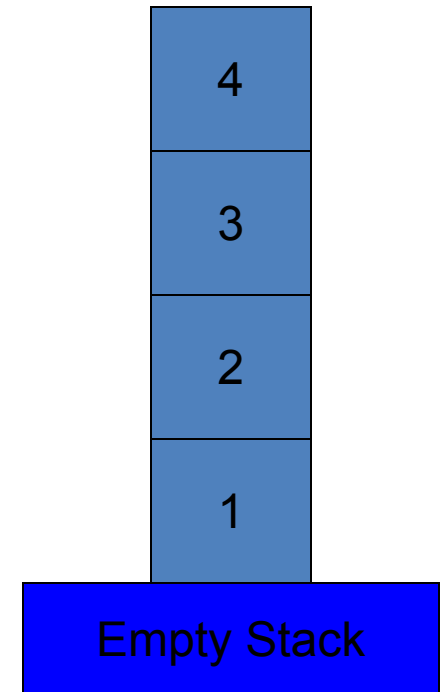
Stack

Stack has only two operation

Push: insert data on top

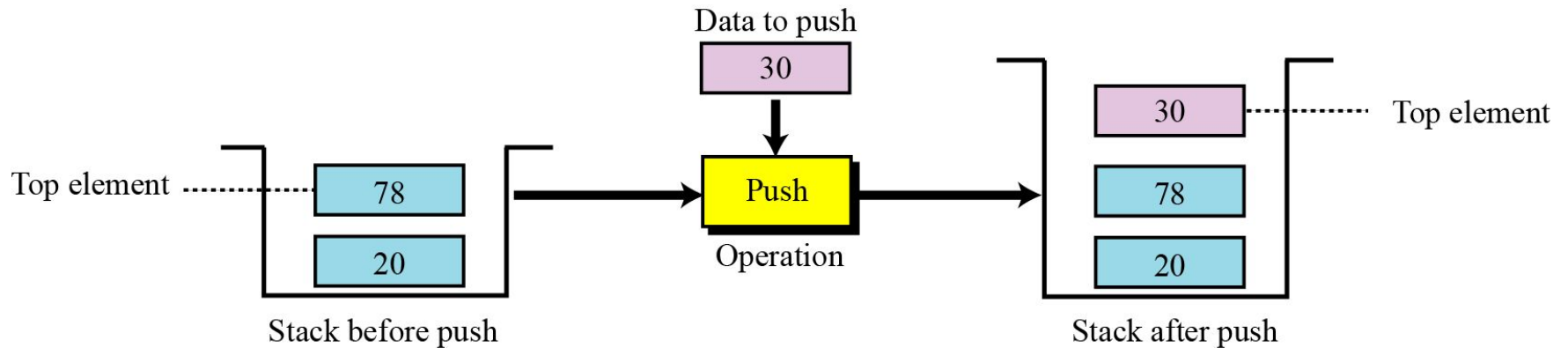
Pop: remove data from top

Also known as LIFO structure
Last In First Out



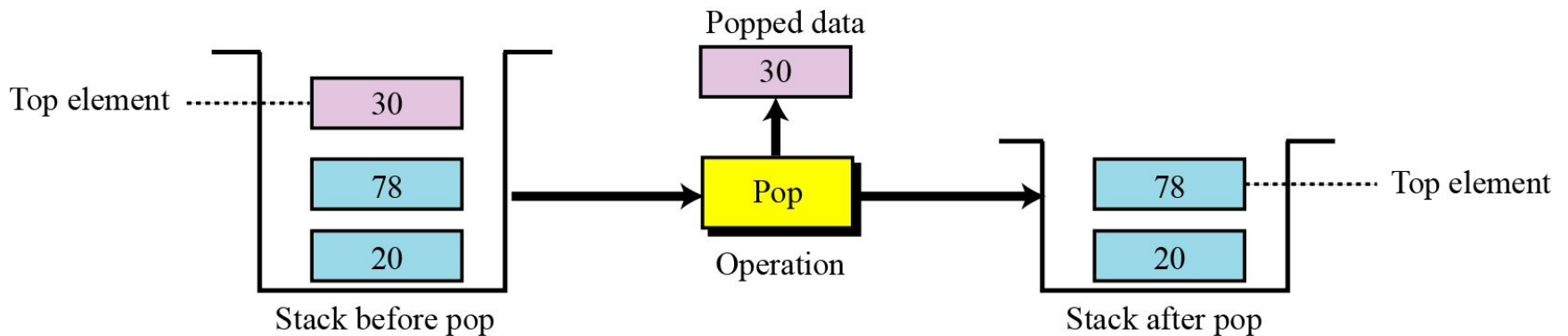
The *push* operation

- The *push* operation inserts an item at the top of the stack. The following shows the format.



The pop operation

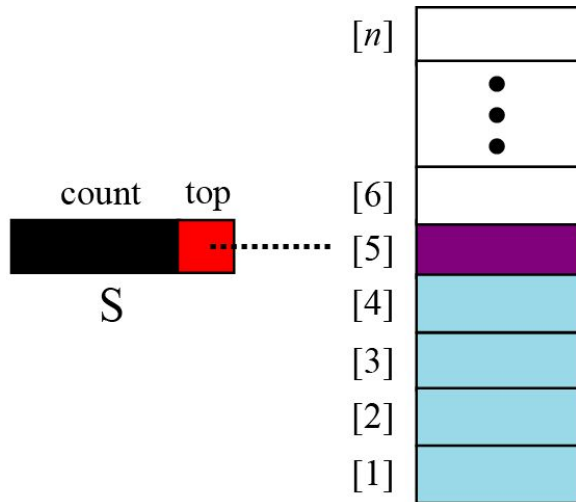
- The **pop operation** deletes the item at the top of the stack. The following shows the format.



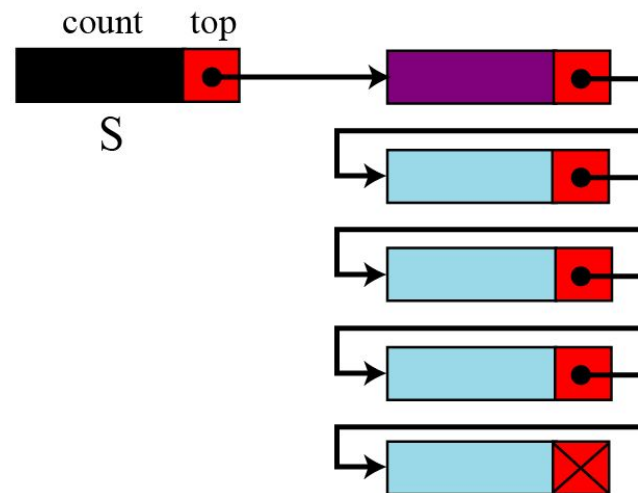
Error checking

- There are two stack errors that can occur:
 - **Underflow**: trying to pop (or peek at) an empty stack
 - **Overflow**: trying to push onto an already full stack

Implementation



b. Array implementation



c. Linked list implementation

Algorithms for Push and Pop

- **Stack is the place** where to store data.
- **Top** represents in which location the data is to be inserted.
- **MaxSTK** is the maximum size of the stack and finally, Item is the new item to be added.

For Push Operation

1. If $Top = MaxSTK$ then Print: Overflow and Return.
/*...Stack already filled..*/
2. Set $Top := Top + 1$
3. Set $Stack[Top] := Item$
4. Return.

For Pop Operation

1. If $Top = -1$ then Print: Underflow and Return.
/*...Stack already Empty..*/
2. Set $Item := Stack[Top]$
3. Set $Top := Top - 1$
4. Return.

Some uses of stacks

- Stacks are used for:
 - Any sort of nesting (such as parentheses)
 - Evaluating arithmetic expressions (and other sorts of expression)
 - Implementing function or method calls
 - Keeping track of previous choices (as in backtracking)